CS & F Case Study Exercise – Possible/Suggested Answers

Exercise 1:

Read through the scenario section "A short overview of the complete production process" only, and identify potential areas where waste could be occurring. Use the "Seven Wastes of Lean" slide as a guide.

Areas of waste to investigate could include…..

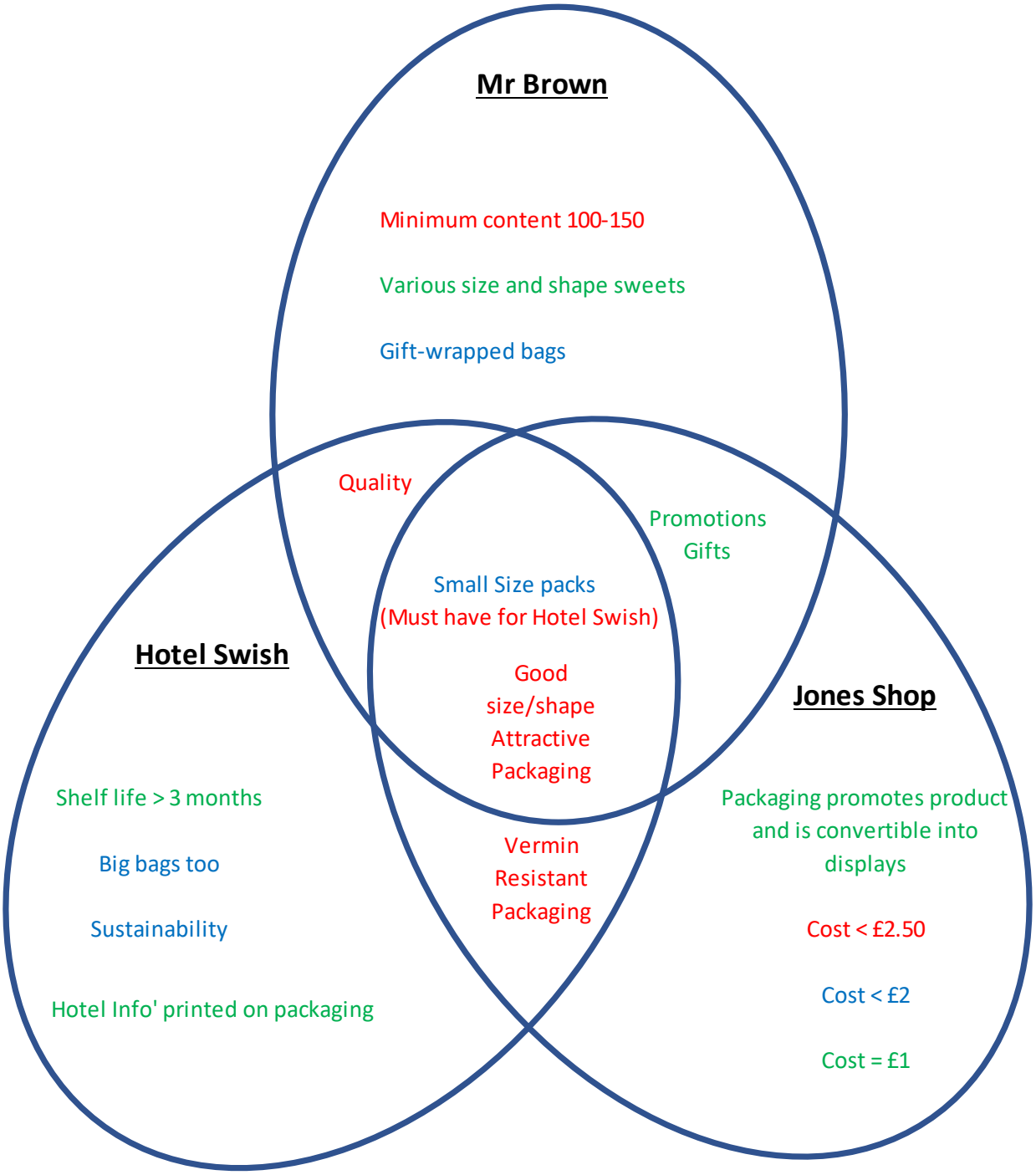
1) Purchasing Sugar in Bulk (in order to get the best price) 'may' be a false economy when you consider the costs of unloading, moving the goods and storing them. It is possible that we are introducing wastes of 'Transportation', 'Motion', 'Inventory', 'Over-Processing' (double handling of the same products) – maybe investigate a more 'pull' and 'Just-in-time' system to reduce ordering and inventory levels to create more of a 'flow' in materials transfer/use.
2) Purchasing flavourings for the 'Target' sweets – if the cost impact is low, why waste so much time and effort in sourcing the cheapest supplier? The time and effort associated with this activity is itself a 'cost' to the business.
3) Testing of the ingredients is done twice. Is this necessary (perhaps due to H&S regs or due to the shelf-life of the product) or is this a potential waste?
4) Purchasing 200 tons of sugar (to obtain a bulk discount) when only 10-12 tons are used per day means that again, we are possibly introducing additional motion, transportation and inventory wastes. The sugar not needed immediately has to be moved into storage and kept there until required. The extra handling and time in storage 'may' also be introducing damage or risk goods passing expiry date (if applicable) – which is also 'waste'!
5) The flavourings cycling between 15 and 45 days in the warehouse makes it very difficult to schedule inventory efficiently (and allocate appropriate storage space) – 'Discussion's with suppliers are needed here to create a more reliable delivery system.
6) The process of heating and melting the ingredients and then dropping them on to a belt is said to be very slow – this could well be a bottleneck in the whole process and probably means this step is much slower than subsequent steps. This impacts the 'flow' of the whole process and creates the 'Waiting' waste as later steps are forced to go at the pace of this step.
7) Having all the vats drop their sweets onto one 'belt' is forcing CS & F to create one product at a time, even though they have multiple vats which could each create a different product. It is worth investigating adjusting the system to introduce multiple belts, although cost and impact may make this impractical.
8) Cartons will spend a lot of time sitting and 'waiting' for the 500 bags necessary to fill each one. Would smaller cartons increase 'flow'?
9) Cleaning time required in order to swap sweet types is obviously necessary, but 2 hours is a large changeover time and is creating the 'waiting' waste again – could this be improved?
10) The 'batch' production mentality that number 9 has instigated results in 3 months worth of inventory (and the motion, transportation, inventory, over-processing (from double handling of goods) wastes are all created as a result).
11) The 'delivery by a variety of methods' could be raising the issues of additional more complicated delivery processes than may be required. Could we not just have one delivery method (plus a contingency)? – worth investigating
12) Cartons waiting at the end of the production line for up to 4 hours before being transferred to despatch is 'waiting' (and 'inventory') and needs to be looked into.
13) The purchasing method currently used by the web team means they can have partially used cartons lying around for up to a week (inventory)
14) The Web Fulfilment Team being treated as a separate entity is creating additional processes, 3rd party contracts (with transport company) and no doubt slows the whole process down significantly. This is likely to be costing the company significantly and should be considered as a priority for IT.

Exercise 2:

Activity 1: Customer requirements for each Archetype could include…

Basic = Red Expected = Blue Exciters = Green

Mr Brown	Hotel Swish	Jones Corner Shop
Cost - as negligible disposable income	Quality – luxurious appearance	Quality
Small bags	Small bags for rooms	Attractive packaging – impulse buys
Good Quality	Big bags for reception bowl	Shape of Packaging – easy placement
Nice Packaging	Sustainability/Fair Trade requirement	Size – easy to pick up
Minimum content = 100-150	Shaping of Packaging – for 'Pillow's and storage	Promotional (2 for 1) ideas and 'gifts
Non-uniformity of sweets would be nice	Resistant packaging – vermin	Packaging promotes product / easily placed in 'display' format
Gift wrapped	Hotel details printed on packaging	Resistant packaging (mice)
Surprise 'gift' inside (competition?)	Decent shelf life (> 3 months)	Price = must be < £2.50. Good to be < £2.00. brilliant to be £1



Activity 2: Affinity Diagram could include ……

= Mr Brown

= Hotel Swish

= Jones Corner Shop

COST	PACKAGING	QUALITY	PROMOTIONS / GIFTS
"Money is tight"	Small bags	"Good quality"	Surprise gift inside - i.e Competition
"A £2 bag ……" (inferring preferred price point)	"Nice" packaging	Various sweet shapes and sizes	Gifts/Promotions i.e 2 for 1
Price MUST be < £2.50 Would Like < £2.00 Great to be < £1.00	Gift wrapped	Minimum content of 100 Preferred content 150	
	Small bags for hotel rooms	Luxurious Appearance	
	Big bags for reception bowl	Sustainability / Fair Trade requirement	
	Vermin resistant packaging	Shelf Life > 3 months	
	Hotel details printed on packaging	Decent Quality	
	Correct shape to lay on pillows		
	Attractive packaging - impulse buying		
	Good shape to place on counter		
	Small size / shape - easy to pick up		
	Packaging promotes product - easily placed into a 'Display' format		
	Vermin resistant packaging		

GROUP DESCRIPTIONS:

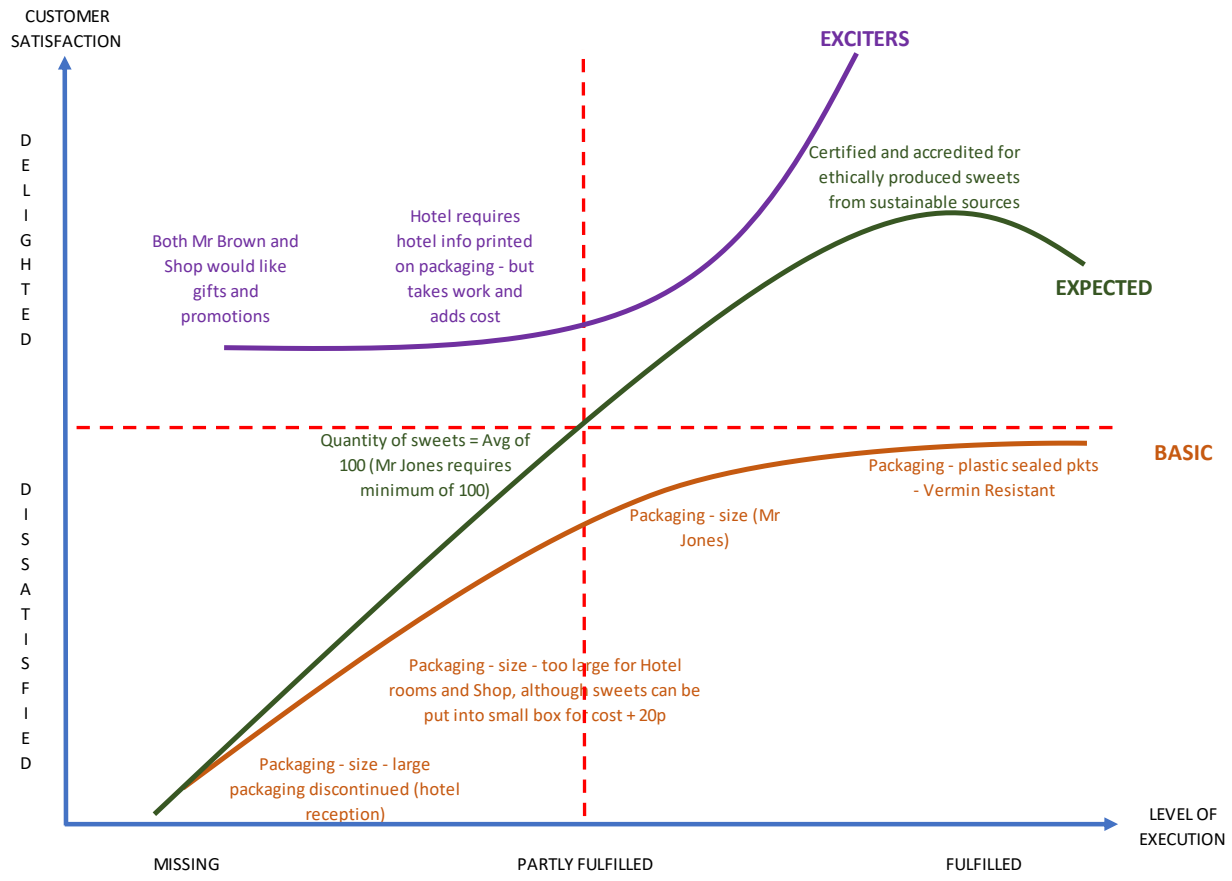
COST - 2 of the 3 customer archetypes state cost as an influencing factor, with Mr Jones having limited funds and implying that a typical bag of sweets containing between 100 and 150 sweets should cost around £2. Jones Corner shop seems to be agreeing with the preferred £2 price point with a maximum price of £2.50 and an ideal price of £1. Overall could be set as an 'EXPECTED' level group (providing cost is < £2.50 which is a 'BASIC' requirement.

PACKAGING – All customer types are consistently quoting numerous requirements concerning packaging and are largely in consensus with these requirements – such as Vermin resistant packaging, small nicely shaped for placement and handling etc. As a number of these requirements are 'Needed', overall this group would most likely be classed as 'BASIC'

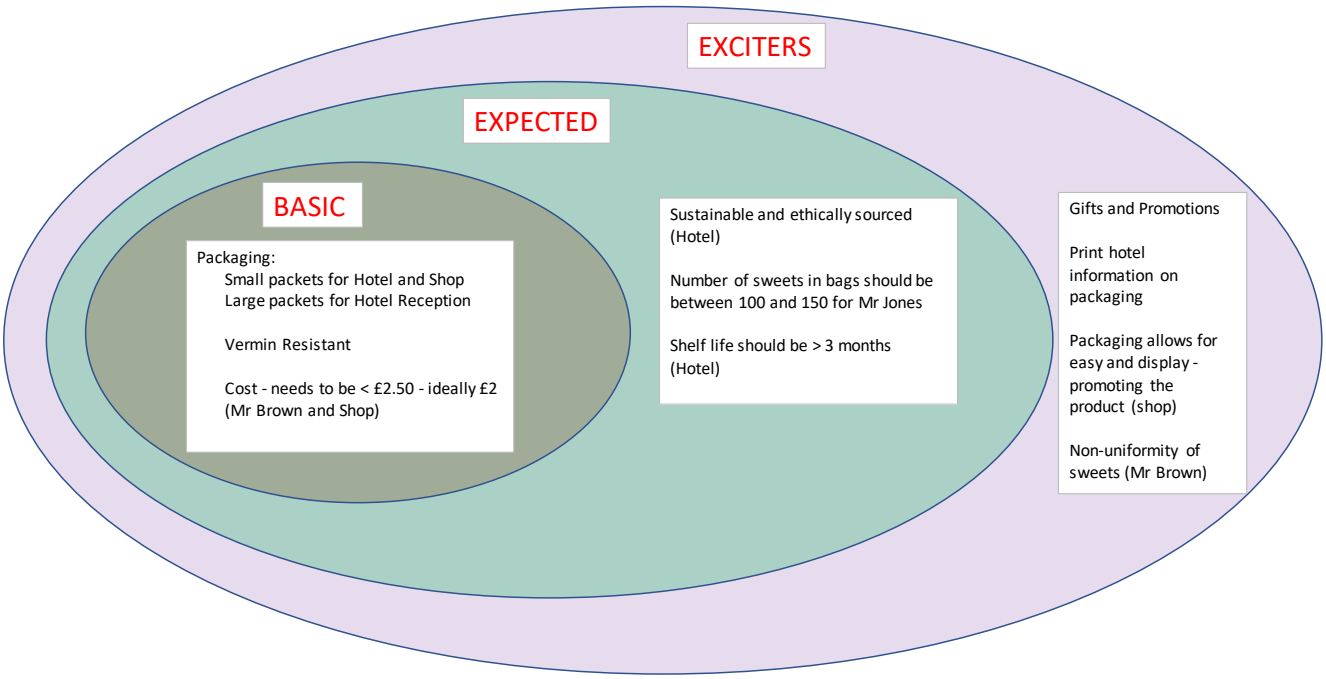
QUALITY – Mostly a requirement of the Mr Brown and Hotel Swish customer types (although there is a generic requirement about good quality from Jones corner shop), with specific quality criteria being stated (such as luxurious appearance, minimum content, varied shape, minimum shelf life, meeting sustainability and fair trade requirements etc). As this is an important and fundamental requirement for Mr Jones, overall, this group should be labelled as 'BASIC'

PROMOTIONALS/GIFTS – Both Mr Jones and Jones Corner Shop would like to see some form of promotional offer or gift idea included in sweet bags. Overall this could be set as 'EXCITERS' as it seems to be a nice to have, rather than a necessary requirement.

Exercise 3: KANO diagram could include ……



Alternatively – a Feature Summary Diagram could include...



Exercise 3: SIPOC

EXAMPLE SIPOC

(items in black are from scenario; items in green are possible additional considerations)

SUPPLIER(s)	INPUT(s)	PROCESS STEP	OUTPUT(s)	CUSTOMER(s)
Stock inventory system Warehouse team	Warning Flag from stock inventory system - either via automatic notification or from manual monitoring by warehouse team ?	Warehouse send email to Purchasing	Email Notification Updated stock control system	Purchasing Team
Warehouse team Purchasing team IT (or Web Portal supplier) CS&F manufacturing dept FSA (or QA/Governance dept) Supplier Database	Email Request Available person to do tender B2B Web Portal Quantity and other reqmt's List of legal requirements Preferred supplier list	Purchasing log on and place tender, specifying amount and requirements	Tender document (including all relevant details and delivery date) Updated systems Warehouse updated	All eligible suppliers IT systems Warehouse team
Purchasing/Tender system Interested suppliers Purchasing Dept / systems Purchasing team Purchasing policies / SOP's	Alert that 2-3 days is up Receivd Bids Order forms / system Bid assessor / evaluater Bid assessment criteria / weighting policies etc	Tender closed and winning bid accepted	Order Evidence of fairly evaluated bid Updated systems / advice to Warehouse that order placed and expected delivery date Notification to update Finance system	Winning supplier QA / Internal governance Purchasing system Warehouse Finance
Winning Supplier Warehouse team Quality Control team Quality Control / Governance / FSA	Delivered stock Stock 'handlers' to unload and move to correct location Quality inspector(s) Quality inspection criteria and required regulations etc	Order received and tested	Accepted stock Quality insepection Certificate Non-acceptable stock Notification to Purchasing to either pay invoice or contact supplier for replacement	Warehouse QA / Purchasing / Warehouse Warehouse to hold pending collection or return Purchasing
Quality inspector	Notification that delivery passed or failed	Purchasing either contact supplier for replacement order or notify Finance to pay invoice	Notification to Finance to pay or Replacement order placed Systems updated	Finance Supplier Warehouse systems Purchasing systems
Purchasing Finance Dept	Instruction to pay suppliers invoice Company guidelines on paying invoices	Finance pay invoice	Notification sent to supplier and purchasing to advise payment made P.O raised to pay invoice Funds Transferred Systems updated	Supplier Purchasing Purchasing Mgr (signatory) Supplier Finance & Purchasing systems

Exercise 4: Risk

EXTRACT RISK REGISTER

RISK ID	DESCRIPTION	PROBABILITY	IMPACT	PROXIMITY	DETECTABILITY	EARLY WARNING INDICATOR (EWI)	MITIGATION RESPONSE	SECONDARY RISKS	RESIDUAL PROBABILITY	RESIDUAL IMPACT
1	H & S risk of installer falling through the roof whilst installing the sensor	Low	V.High (risk to life, legal issues, compensation , reputational damage)	< 3 mths	Low	Secondary person assisting and monitoring to notice difficulties or boards coming loose etc	REDUCE - wider permanently fixed crawl boards fixed professionally, with small rail guards to reduce chance of slipping off	Increased weight may decrease lifespan of already fragile roof or even directly cause damage	Very low - due to increased safety measures	No change as financial / legal / health and safety impact is still the same should somebody actually have a fall
2	Dust blown up on windy days could reduce the lifespan of the sensor or impact it's functionality	Med	Med - There will be no manual controls should the sensor fail	< 3 mths	Med (Sensor operating erratically)	Regular inspections of sensor unit	Maintenance contract with supplier to service and replace regularly	More foot traffic on 'fragile' roof may reduce lifespan and increase chance of damage/fall	Low - should reduce due to regular maintenance / replacement	Should reduce as regular maintenance will decrease the effects of dust

Exercise 5: Sample Data Collection Plan

Data Collection Plan

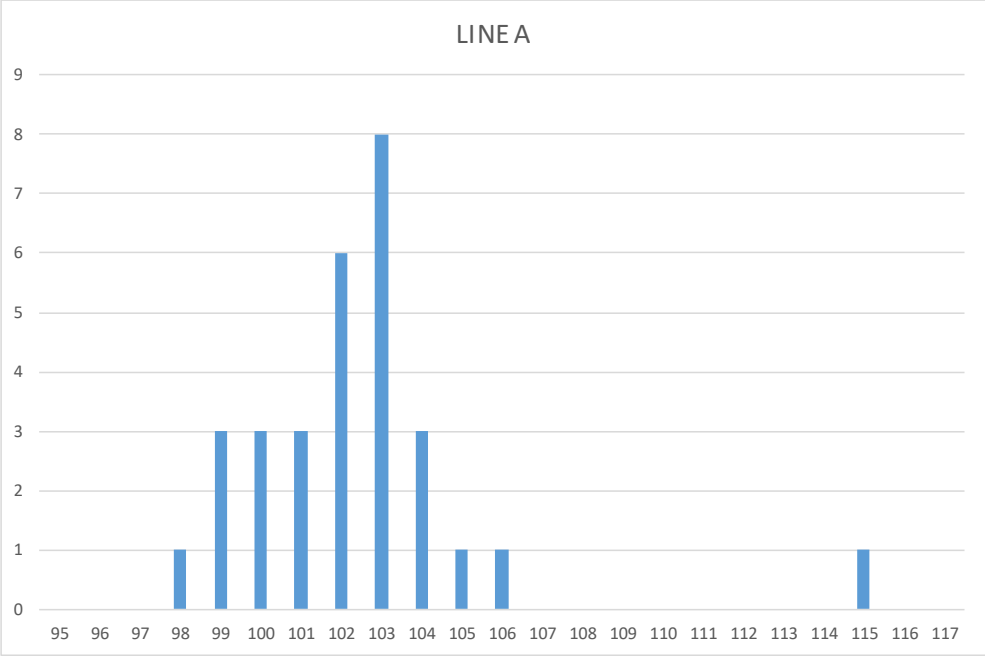
DATA TO COLLECT			MEASUREMENT METHOD		
STATISTIC	HOW RECORDED	DATA TYPE	TIME PERIOD/FREQUENCY	METHOD	PERSON/TEAM RESPONSIBLE
% of incorrectly wrapped sweets	Batch of 100 taken every hour and rounded to nearest 1% Data collected over 1 week	Binary (wrapped correctly or not)	Commencing from 6am at hourly intervals, until 6pm (12 samples per day)	Random sample of 100 sweets removed from production line and assessed for correct wrapping	Member of Wrapping Station Team
% of incorrectly formed sweets (incorrect shapes)	Batch of 100 taken every hour Round to nearest 1% Data collected over 1 week	Binary (Correctly formed or not)	Commencing from 6am at hourly intervals, until 6pm (12 samples per day)	Random sample of 100 sweets removed from production line and assessed for correct shape	Member of Production team
Average number of sweets in bag	Content of 10 bags counted every hour Data collected over 1 week	Discrete (counting number of sweets in a bag)	Taken at hourly intervals from 6am until 6pm (12 samples per day)	10 bags randomly selected, contents counted and an average taken over the 10 bags	Bagging station supervisor

Exercise 6: Production Problems

ACTIVITY 1 : Histograms

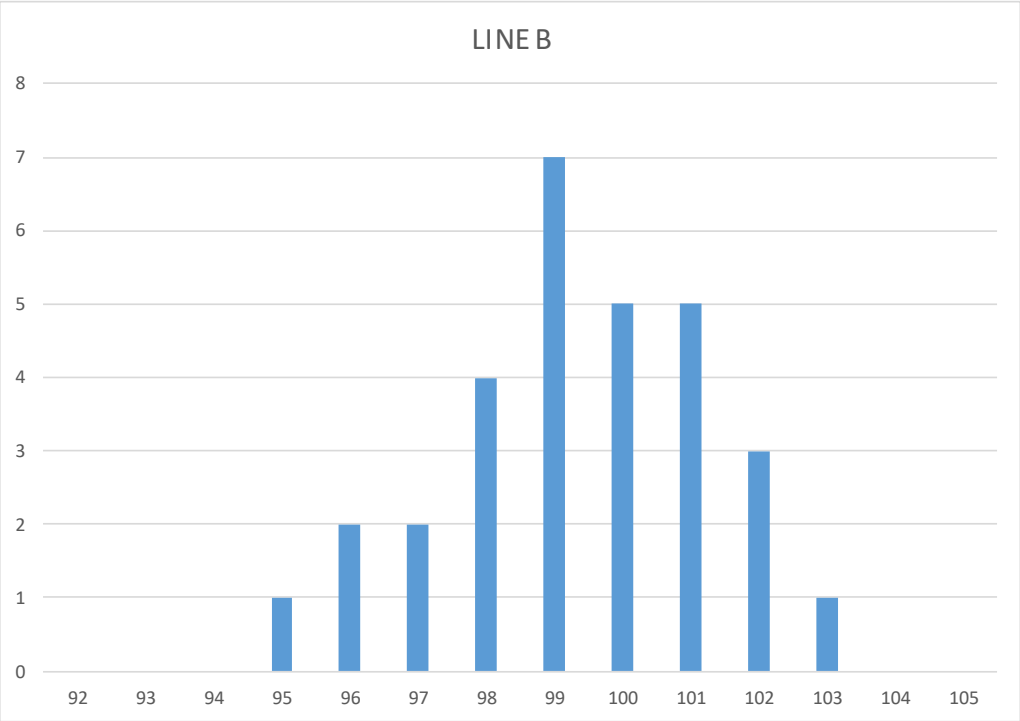
LINE A

Contents	Frequency
95	0
96	0
97	0
98	1
99	3
100	3
101	3
102	6
103	8
104	3
105	1
106	1
107	0
108	0
109	0
110	0
111	0
112	0
113	0
114	0
115	1
116	0
117	0



LINE B

Contents	Frequency
92	0
93	0
94	0
95	1
96	2
97	2
98	4
99	7
100	5
101	5
102	3
103	1
104	0
105	0



ACTIVITY 2 : Control Charts

Voice of the Customer

Current Trading Standards law mandates that sweet bags showing a nominal content of 100 must contain at least 98 sweets

Management want to restrict overfill to a maximum of 2 sweets in order to contain costs.

For both production lines use a SD of two sweets.
Mean (BAR-X) for Production Line A = 102 sweets
Mean (BAR-X) for Production Line B = 99 sweets

LINE A

101	103	102	103	115	102
101	103	103	103	99	102
100	101	100	105	103	102
99	104	102	104	99	100
104	103	98	102	103	106

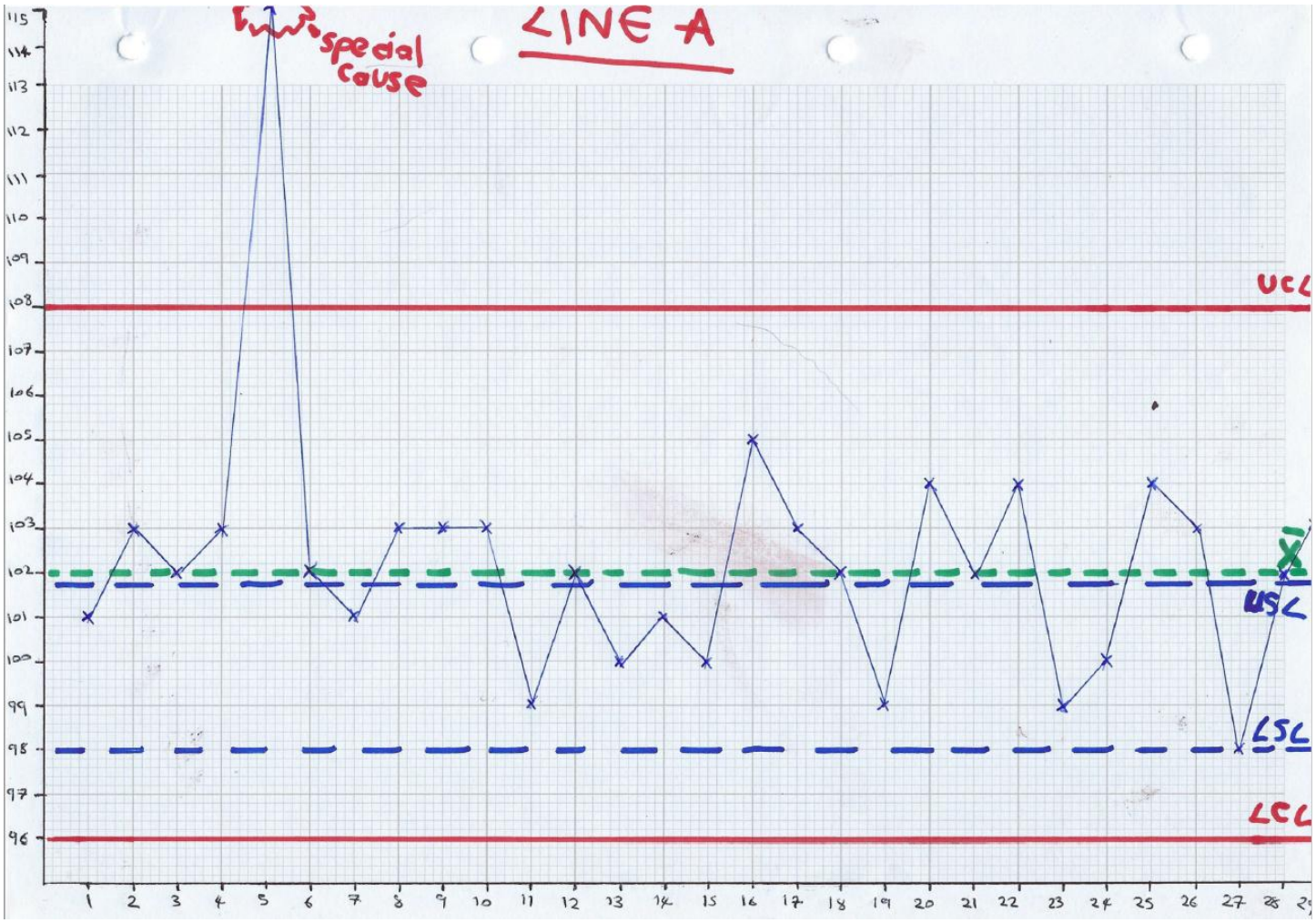
Mean = 102

SD = 2 Therefore 3 x SD = 6

UCL = Mean + 3xSD = 102 + 6 = 108

LCL = Mean - 3xSD = 102 - 6 = 96

Highest value = 115 / Lowest value = 96 (LCL)



LINE B

99	101	96	98	99	100
95	99	103	102	101	100
99	98	97	96	98	102
99	101	101	100	99	102
100	99	100	98	101	97

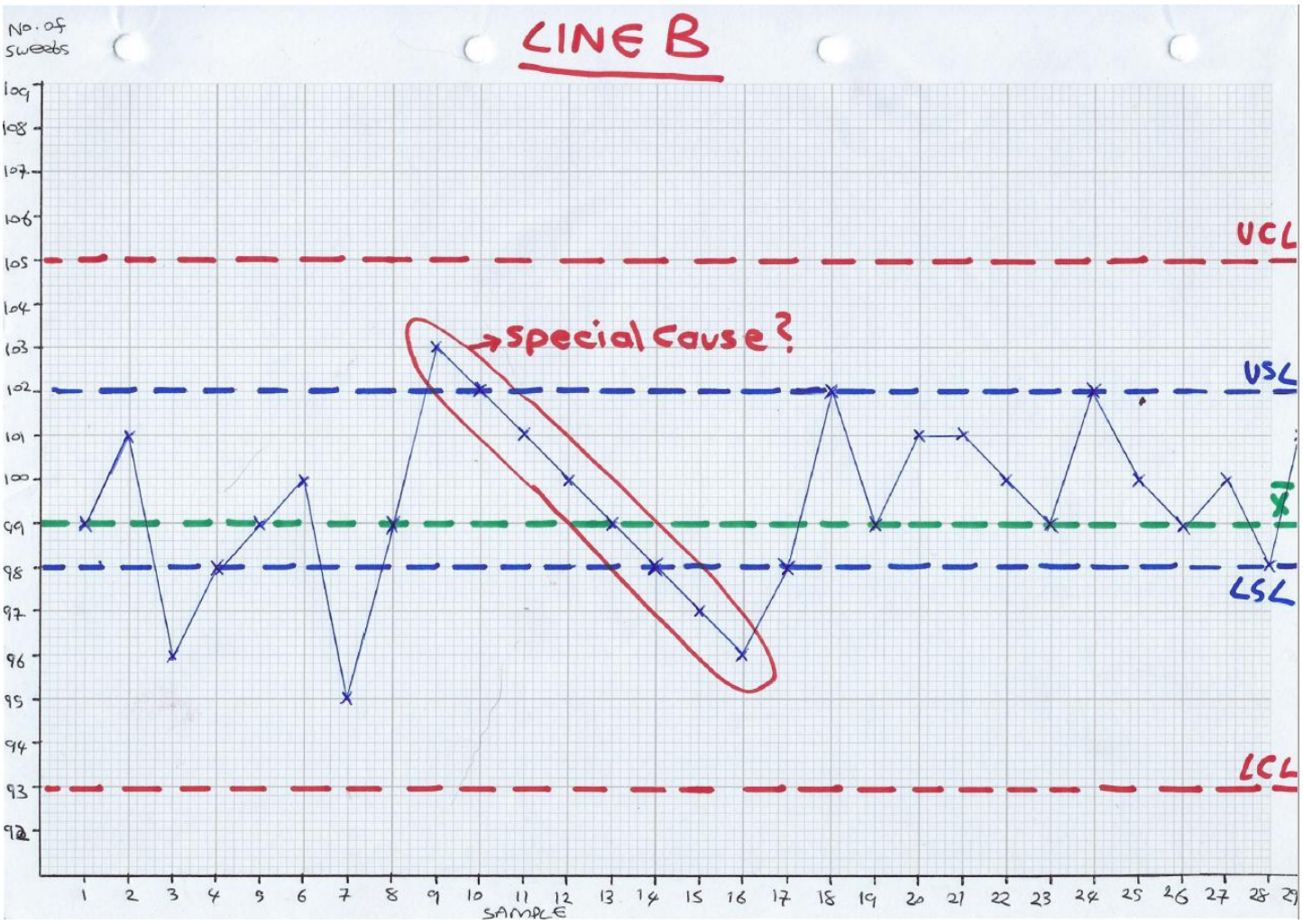
Mean = 99

SD = 2 Therefore 3 x SD = 6

UCL = Mean + 3xSD = 99 + 6 = 105

LCL = Mean + 3xSD = 99 - 6 = 93

Highest value = 105 (UCL) / Lowest value = 93 (LCL)



ACTIVITY 3 : DPMO

As part of their quality checking, the Compliance team monitored the defects on Production Line B.
The team took a sample of 1000 sweets and assessed them against the criteria of:

- Wholly wrapped
- Acceptable shape
- Correct weight

They found that a total of 8 sweets were outside requirements as follows:

SWEET	PARTIALLY WRAPPED	UNACCEPTABLE SHAPE	TOO HEAVY	TOO LIGHT
1	x	x		
2		x		
3			x	
4			x	
5		x		x
6			x	
7	x			
8	x	x		

Using the above information as required, calculate the DPMO and the Long Term Sigma level of this process

ANSWER

$$\text{DPMO} = \frac{D}{N \times O} \times 1{,}000{,}000$$

D = no' of Defects (NOT Defectives) = 11 (number of x's in table)

N = 1000 (number of sweets in batch)

O = 3 (criteria which sweets will be assessed against - taken from question - NOT the table !!!)

So..... $\frac{11}{1000 \times 3} \times 1{,}000{,}000$ gives us $\frac{11}{3000} \times 1{,}000{,}000$

gives us 0.003666 x 1,000,000

DPMO = 3,666.67

Using Conversion Table, Long Term Sigma Value is between 2.6 and 2.7

ACTIVITY 4 : Scatter Diagrams

The Quality Improvement Team is considering two hypotheses for possible root cause(s) of the variation in bag fill. The two hypotheses which have been identified are:

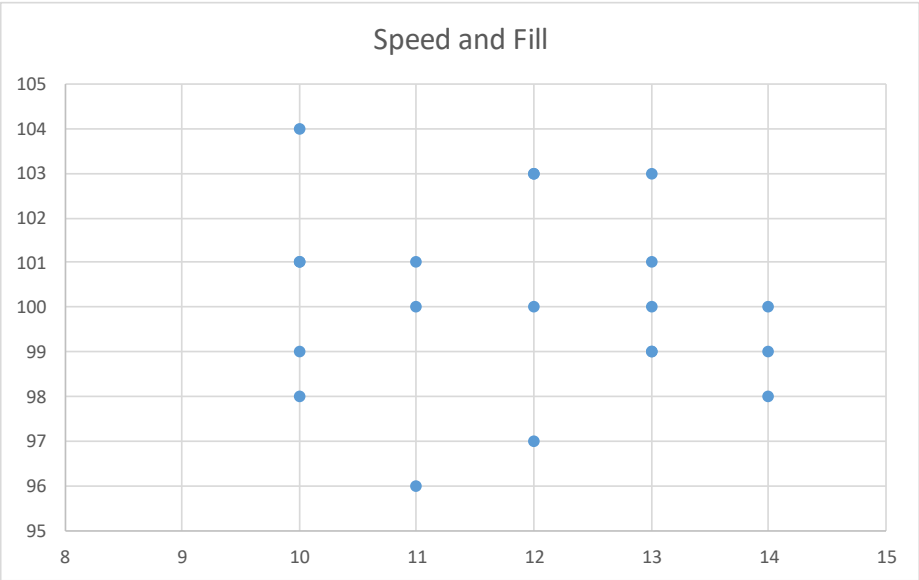
1. Changes in the speed of the production line are causing the variation in bag fill

2. Changes in the weight of the sweets is causing a variation in bag fill

A data collection plan is prepared and the following samples taken:

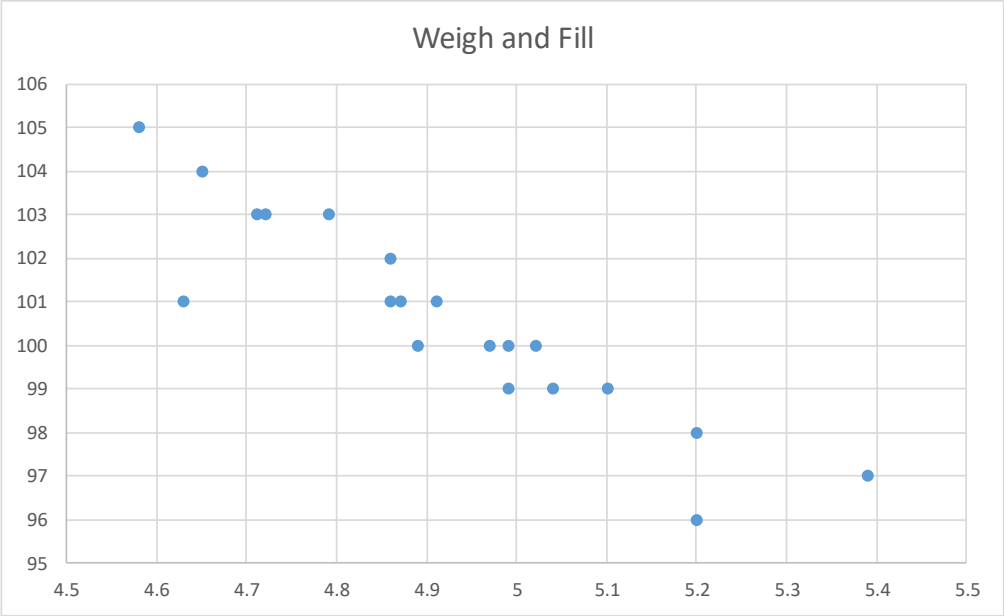
1. Changes in the speed of the production line are causing the variation in bag fill

Speed (mm/s)	Fill
10	101
12	97
13	100
13	99
10	104
10	98
10	101
11	96
11	100
12	100
12	103
12	103
13	101
13	99
13	103
14	99
14	100
14	98
10	99
11	101



2. Changes in the weight of the sweets is causing a variation in bag fill

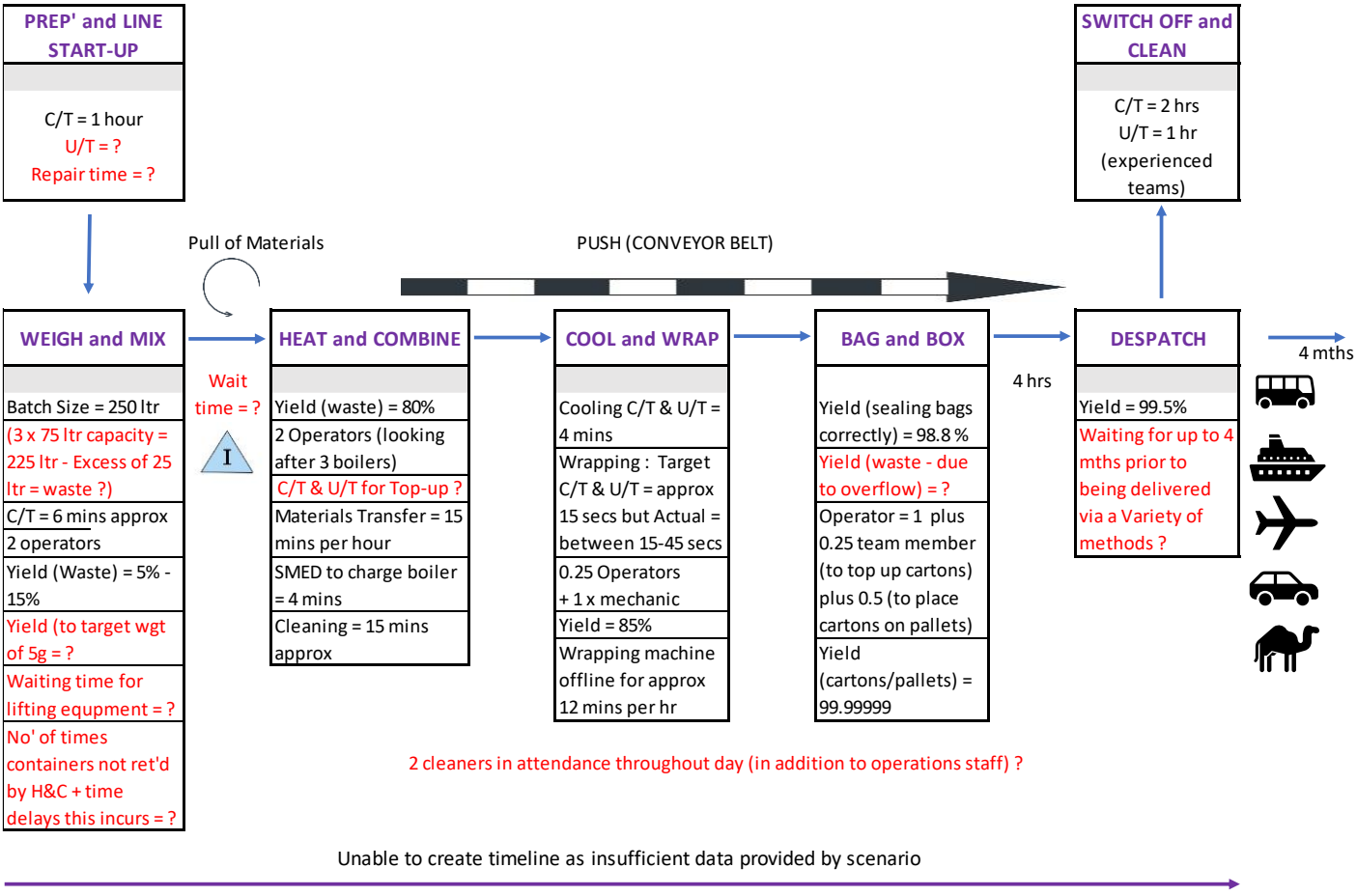
Weight	Fill
4.87	101
4.63	101
4.97	100
5.1	99
4.86	102
5.04	99
5.02	100
5.2	96
4.91	101
4.89	100
4.65	104
4.79	103
5.39	97
4.71	103
4.58	105
4.86	101
5.2	98
4.99	100
4.99	99
4.72	103



Exercise 7: Sample Value Stream Map

Notes in red are where additional measure phase investigation should be done in order to provide actual statistics

PARTIAL VALUE STREAM MAP



Potential areas of waste include:

Weight and Mix
- 25 litres excess mixed for each batch
- Unused waste = 5 - 15%
- Incorrect weight (100g) = ?
- Delays due to waiting for lifting equipment
- Delays due to containers not returned by Heat & Combine

Heat and Combine
- 20% sticks to surface of boilers (plus wasted time spent clearning this off !)

Cool and Wrap
- 15% of sweets incorrectly wrapped and thrown away
- Time - wrapping machine offline 20% of each hour

Bag and Box
- Waste due to overflowing hoppers = ?
- Pallets waiting 4 hours prior to transferring to Despatch warehouse

Despatch
- Goods waitiing up to 4 months prior to shipment ?
- Delivery via variety of methods means overprocessing of sourcing suppliers/contracts etc

Bottleneck(s)

Wrapping Station is most likely bottleneck as it regularly goes offline

Priority

Although more data needs to be collected, the Wrapping station would appear to be having the greatest impact when looking at time and waste